

Lecture 13

Packaging & Power and Clock Networks

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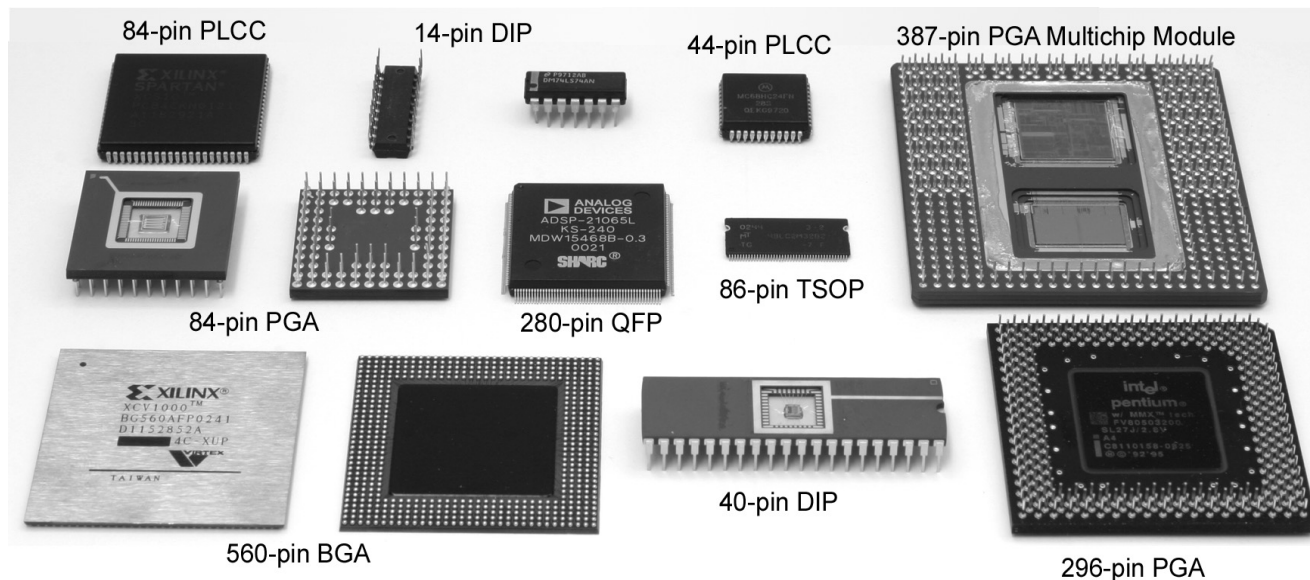
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Packages

◆ Package functions

- Electrical connection of signals and power from chip to board
- Little delay or distortion
- Mechanical connection of chip to board
- Removes heat produced on chip
- Protects chip from mechanical damage
- Compatible with thermal expansion
- Inexpensive to manufacture and test



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Package Options

Package	# I/Os	Description
Dual Inline Package (DIP)	8, 14, 16, 20, 28, 40, 64	Two rows of through-hole pins on 100 mil centers. Low cost. Long wires between chip and corner pins.
Pin Grid Array (PGA)	65–391+	Array of through-hole pins on 100 mil centers. Low thermal resistance and high pin counts.
Small Outline IC (SOIC)	8, 10, 14, 16, 20, 24, 28	Two rows of SMT pins on 50 mil centers. Low cost, good for low-power parts with small pin counts.
Thin Small Outline Package (TSOP)	28–86+	Two rows of SMT pins on 0.5 or 0.8 mm centers in a thin package. Commonly used for DRAMs.
Plastic Leadless Chip Carrier (PLCC)	20, 28, 44, 68, 84	J-shaped SMT pins on all four sides on 50 mil centers. Sturdy leads are convenient for socketing.
Quad Flat Pack (QFP)	44–240	SMT pins on all four sides on 15.7–50 mil centers. High density of I/Os. Available in thin (TQFP) and very thin (VQFP) forms as thin as 1.6 mm.
Ball Grid Array (BGA)	49–2000+	Array of SMT solder balls on underside of package on 15.7–50 mil centers. Extremely high density of I/Os with low parasitics. Requires specialized assembly and inspection equipment to blindly attach to array of pads on printed circuit board.
Land Grid Array (LGA)	Many	Similar to BGA, but with gold-plated pads rather than solder balls. Connects to a socket or pads on the PCB.
Chip Scale Packaging (CSP)	Variable	SMT package no larger than 1.2× the die size. A common form of CSP is the flip-chip, which directly connects to a printed circuit board through solder balls on top metal layer of chip. Even higher I/O density and lower parasitics than BGA. Popular for mobile devices.

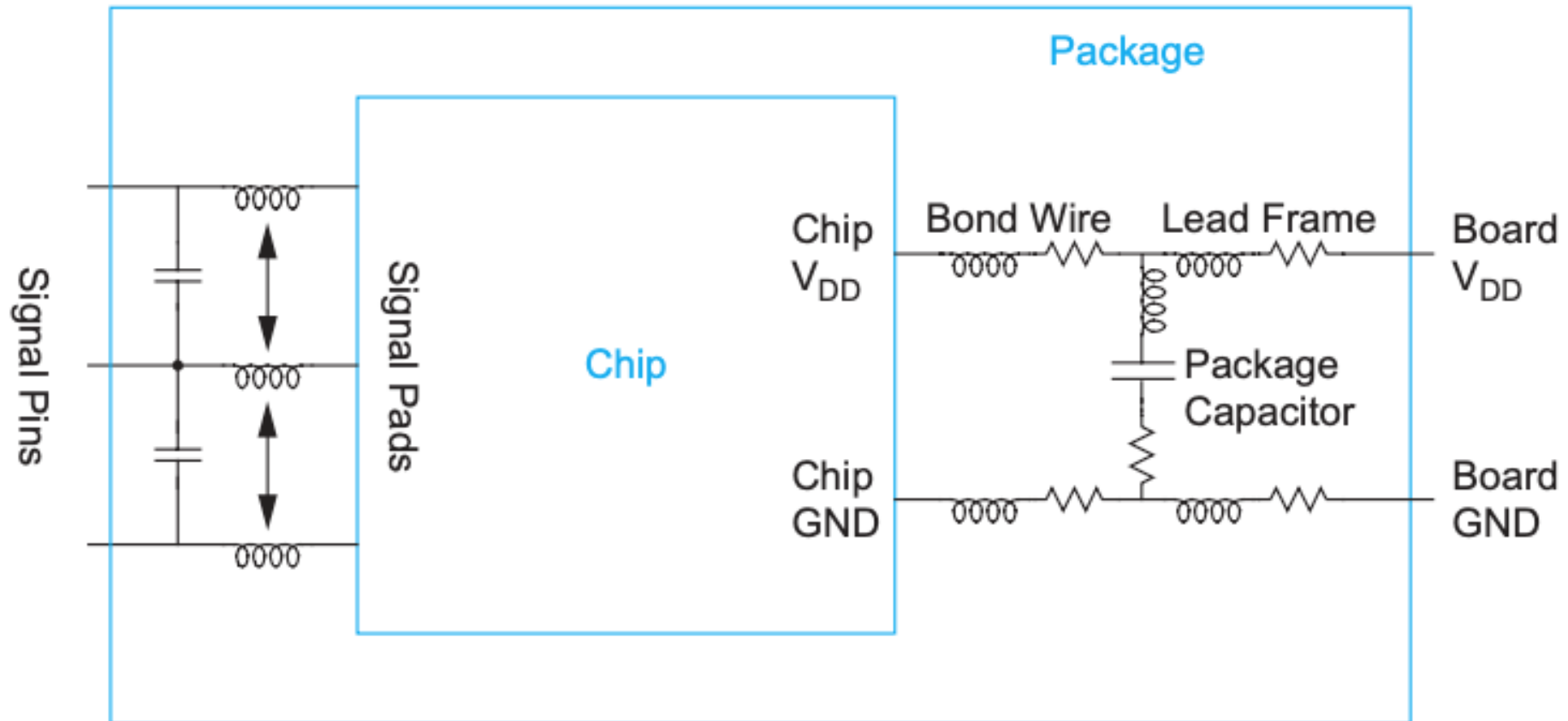
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Advanced Packages

- ◆ Bond wires contribute parasitic inductance
- ◆ Fancy packages have many signal, power layers
 - Like tiny printed circuit boards
- ◆ *Flip-chip* places connections across surface of die rather than around periphery
 - Top level metal pads covered with solder balls
 - Chip flips upside down
 - Carefully aligned to package (done blind!)
 - Heated to melt balls
 - Also called C4 (Controlled Collapse Chip Connection)

Package Parasitics

- ◆ Use many VDD, GND in parallel.



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Heat Dissipation in VLSI

- ◆ 60 W light bulb has surface area of 120 cm²
- ◆ Itanium 2 die dissipates 130 W over 4 cm²
 - Chips have enormous power densities
 - Cooling is a serious challenge
- ◆ Package spreads heat to larger surface area
 - Heat sinks may increase surface area further
 - Fans increase airflow rate over surface area
 - Liquid cooling used in extreme cases (\$\$\$).

◆ $\Delta T = \Theta_{ja} P$

ΔT : temperature rise on chip

Θ_{ja} : thermal resistance of chip junction to ambient

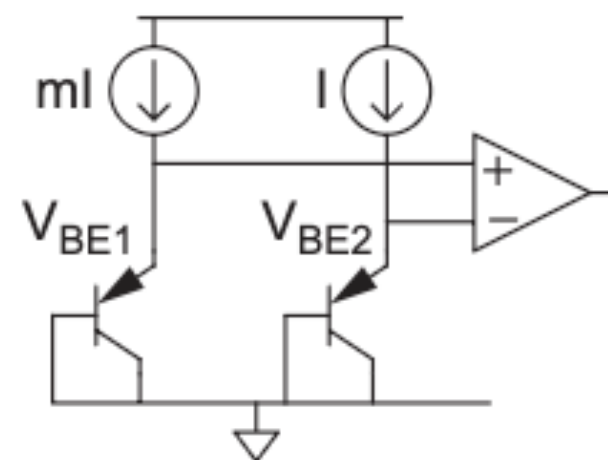
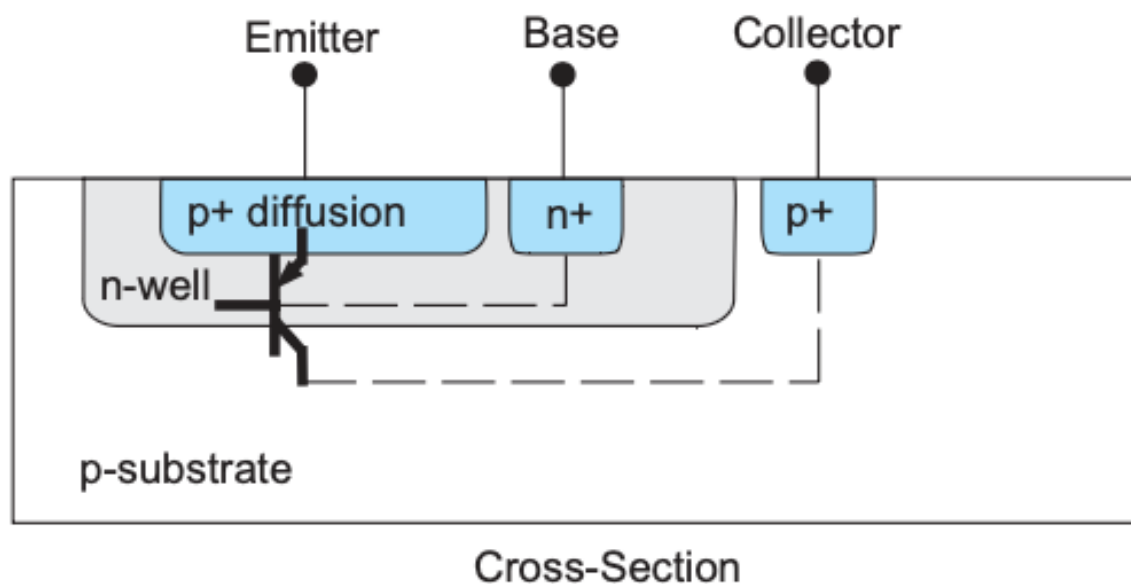
P: power dissipation on chip

- ◆ Thermal resistances combine like resistors: Series and parallel
- ◆ $\Theta_{ja} = \Theta_{jp} + \Theta_{pa}$. (die-to-package + package-to-air)

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On-chip Temperature Sensor (1)

- ◆ Monitor die temperature and throttle performance if it gets too hot.
- ◆ Use a pair of pnp bipolar transistors:
 - Vertical pnp available in CMOS.

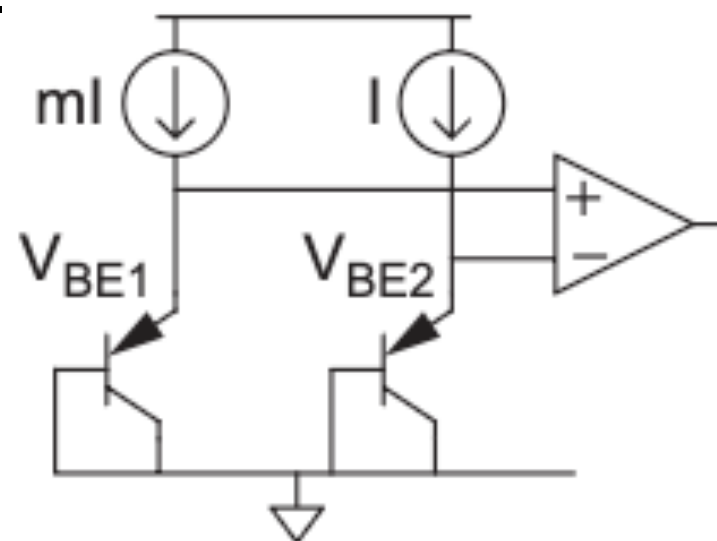


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On-chip Temperature Sensor (2)

- ◆ Voltage difference is proportional to absolute temperature (PTAT).
 - Measure with on-chip A/D converter.

$$I_c = I_s e^{\frac{qV_{BE}}{kT}} \quad \Rightarrow \quad V_{BE} = \frac{kT}{q} \ln \frac{I_c}{I_s}$$



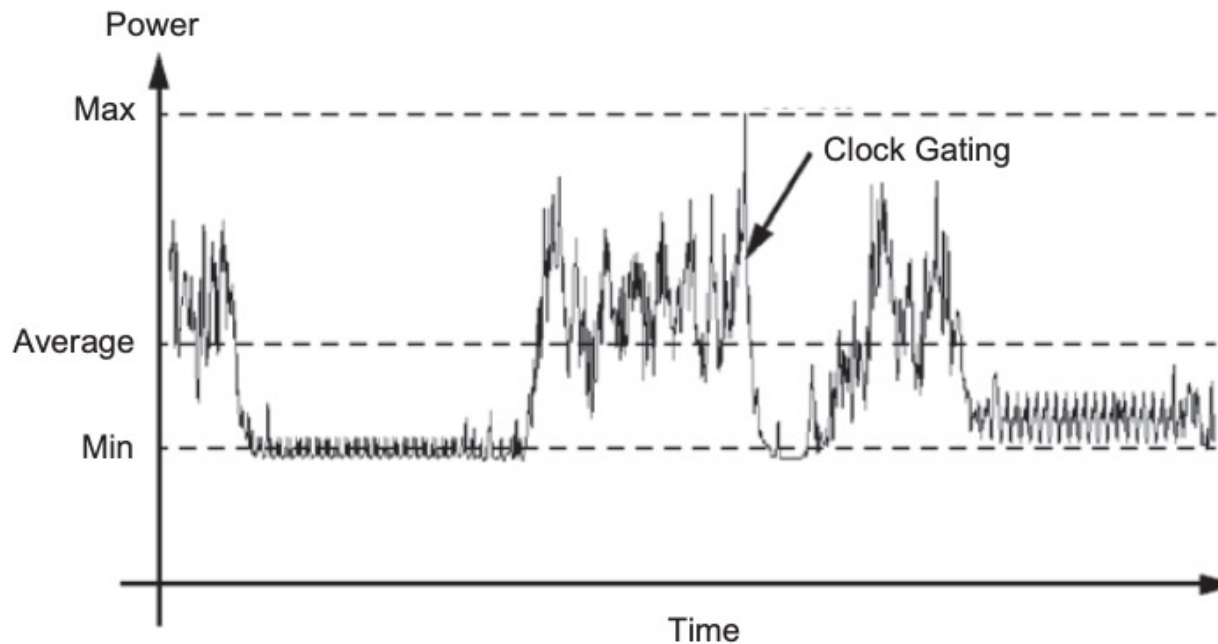
$$\Delta V_{BE} = V_{BE1} - V_{BE2} = \frac{kT}{q} \left(\ln \frac{I_{c1}}{I_s} - \ln \frac{I_{c2}}{I_s} \right) = \frac{kT}{q} \left(\ln \frac{I_{c1}}{I_{c2}} \right)$$

Power Distribution

- ◆ Power Distribution Network functions:
 1. Carry current from pads to transistors on chip
 2. Maintain stable voltage with low noise
 3. Provide average and peak power demands
 4. Provide current return paths for signals
 5. Avoid electromigration & self-heating wearout
 6. Consume little chip area and wire
 7. Easy to lay out.

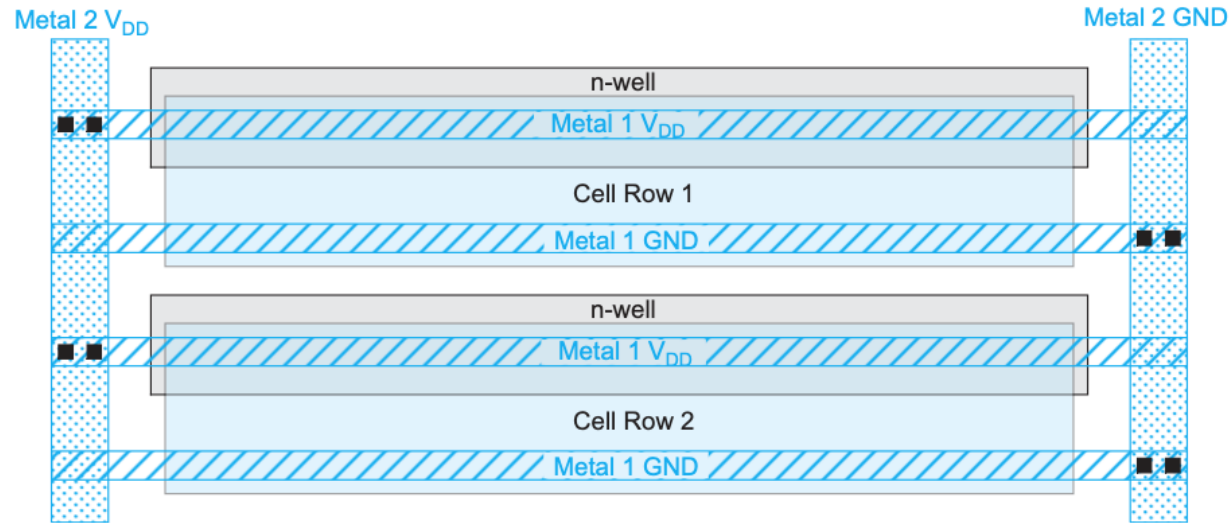
Power Requirements

- ◆ $V_{DD} = V_{DDnominal} - V_{droop}$
- ◆ Want $V_{droop} < +/- 10\%$ of V_{DD}
- ◆ Sources of V_{droop} :
 - IR drops
 - L di/dt noise
- ◆ I_{DD} fluctuates during operation:

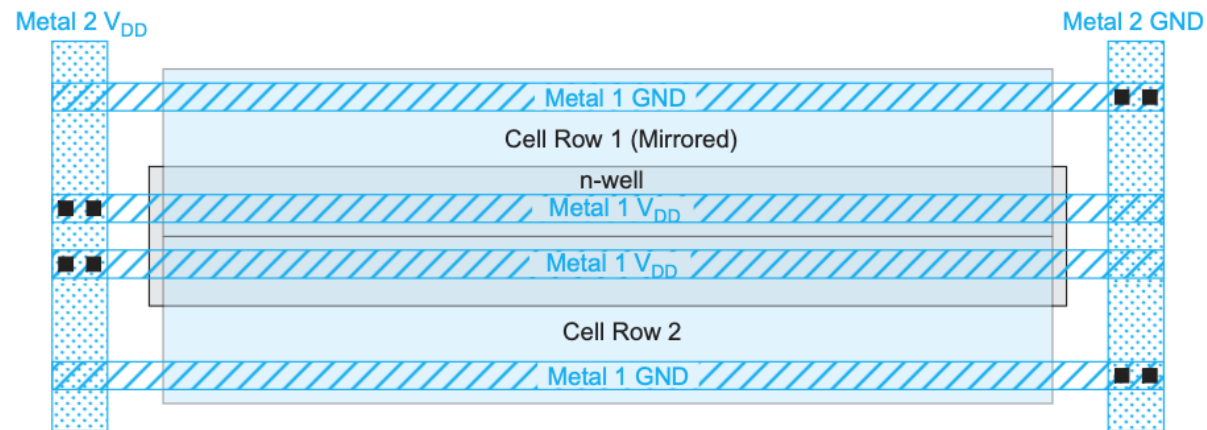


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Power Distribution Network



(a)

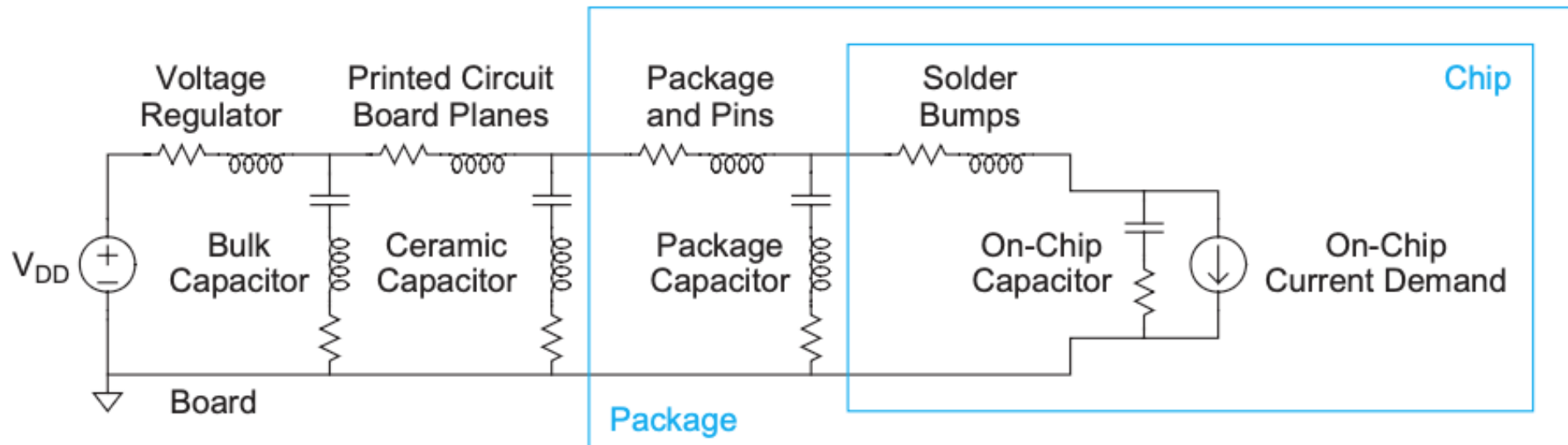


(b)

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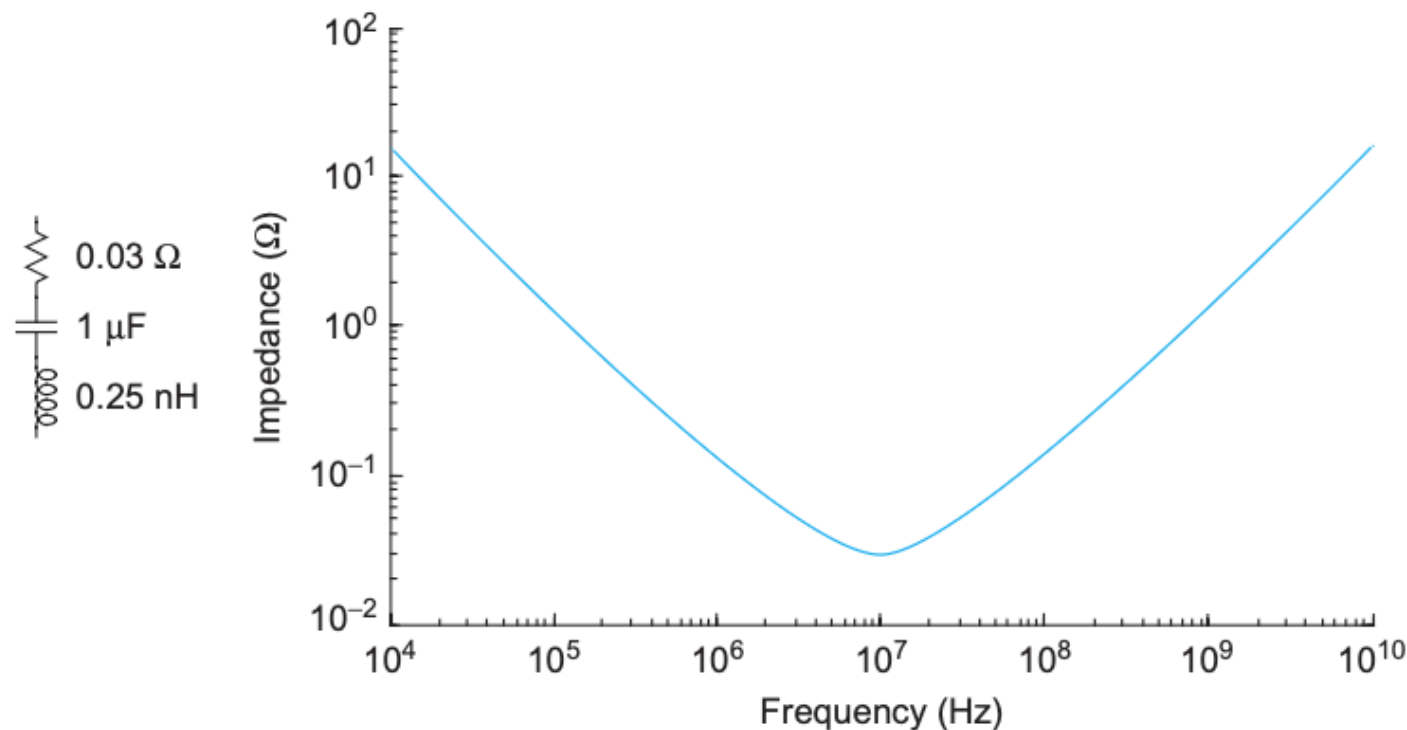
Power Network Model

- ◆ Power comes from regulator on system board:
 - Board and package add parasitic R and L
 - Bypass capacitors help stabilize supply voltage
 - But capacitors also have parasitic R and L
- ◆ Simulate system for time and frequency responses.



Bypass Capacitors

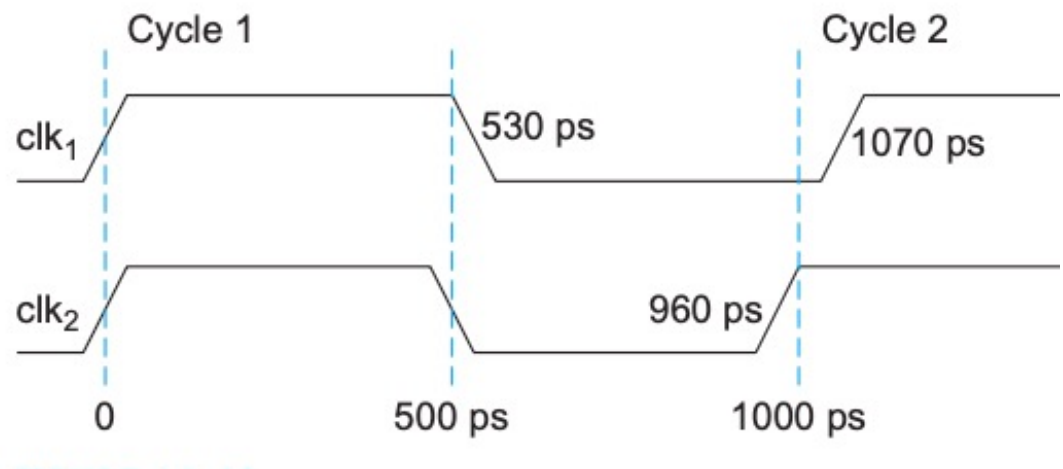
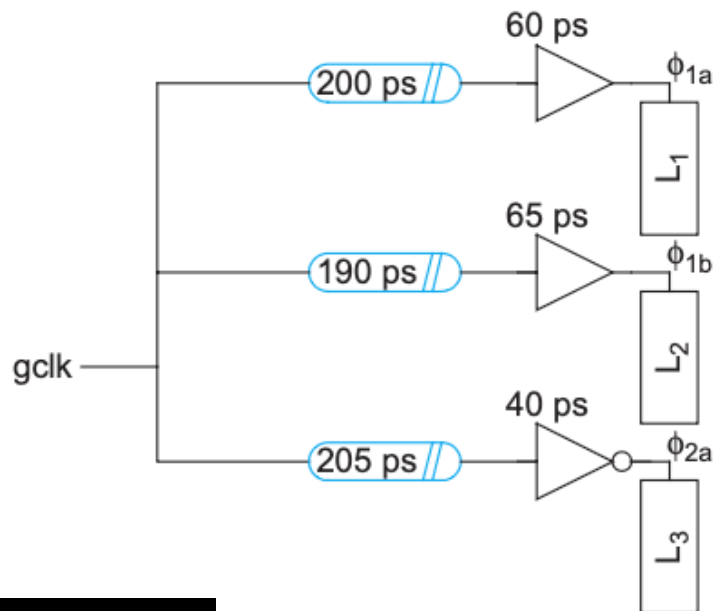
- ◆ Need low supply impedance at all frequencies
- ◆ Ideal capacitors have impedance decreasing with ω
- ◆ Real capacitors have parasitic R and L
 - Leads to resonant frequency of capacitor :



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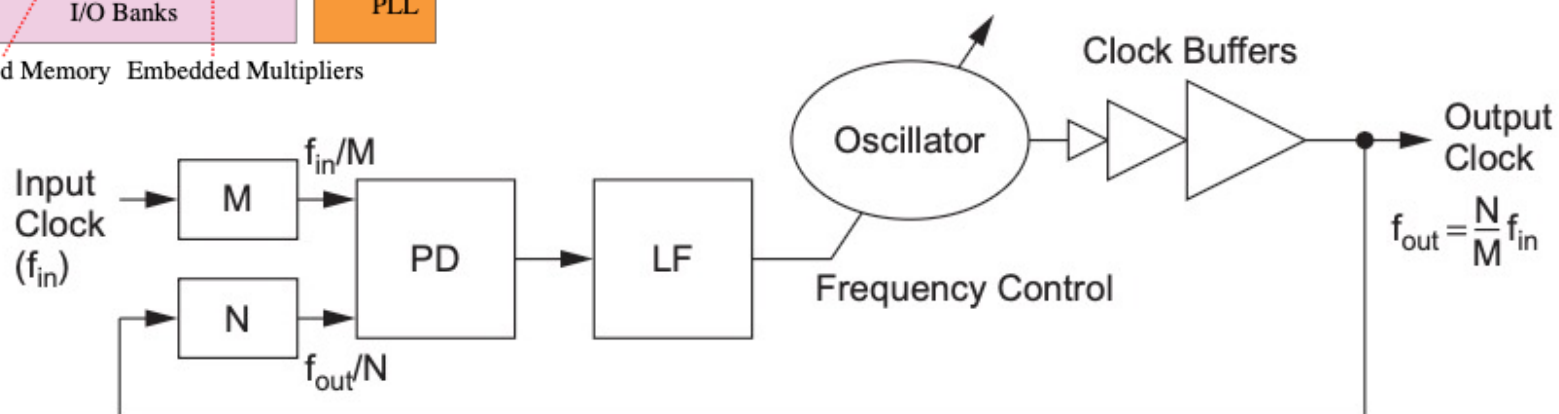
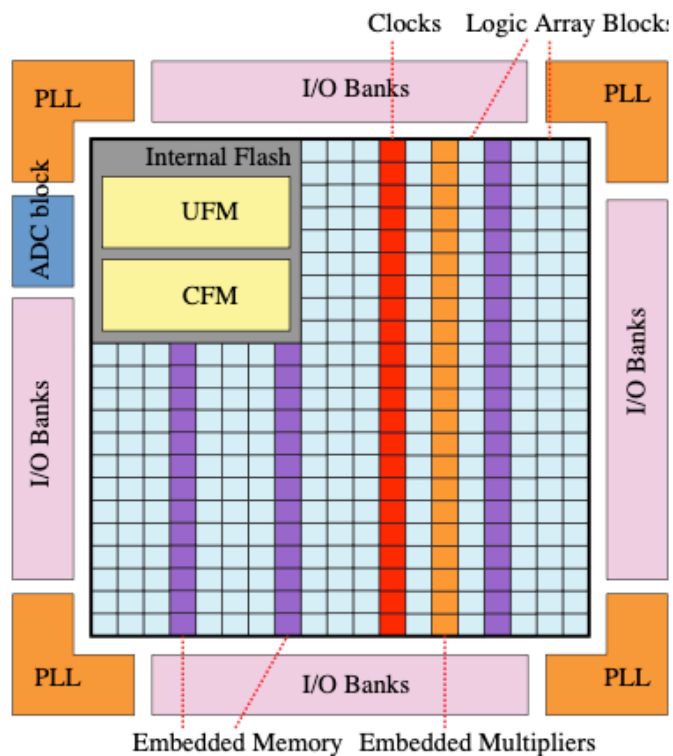
Clock Network

- ◆ Use grid on two or more levels to carry clock
- ◆ Make wires wide to reduce RC delay
- ◆ Ensures low skew between nearby points
- ◆ But possibly large skew across die



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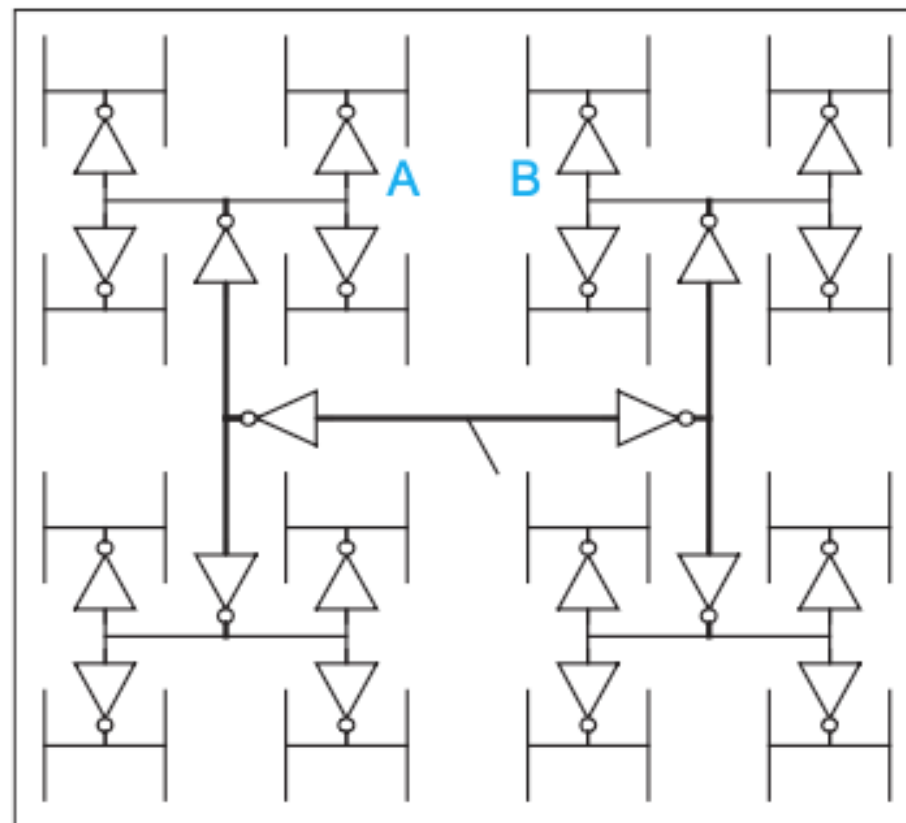
PLL in MAX 10 FPGA



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Clock Distribution using H-Trees

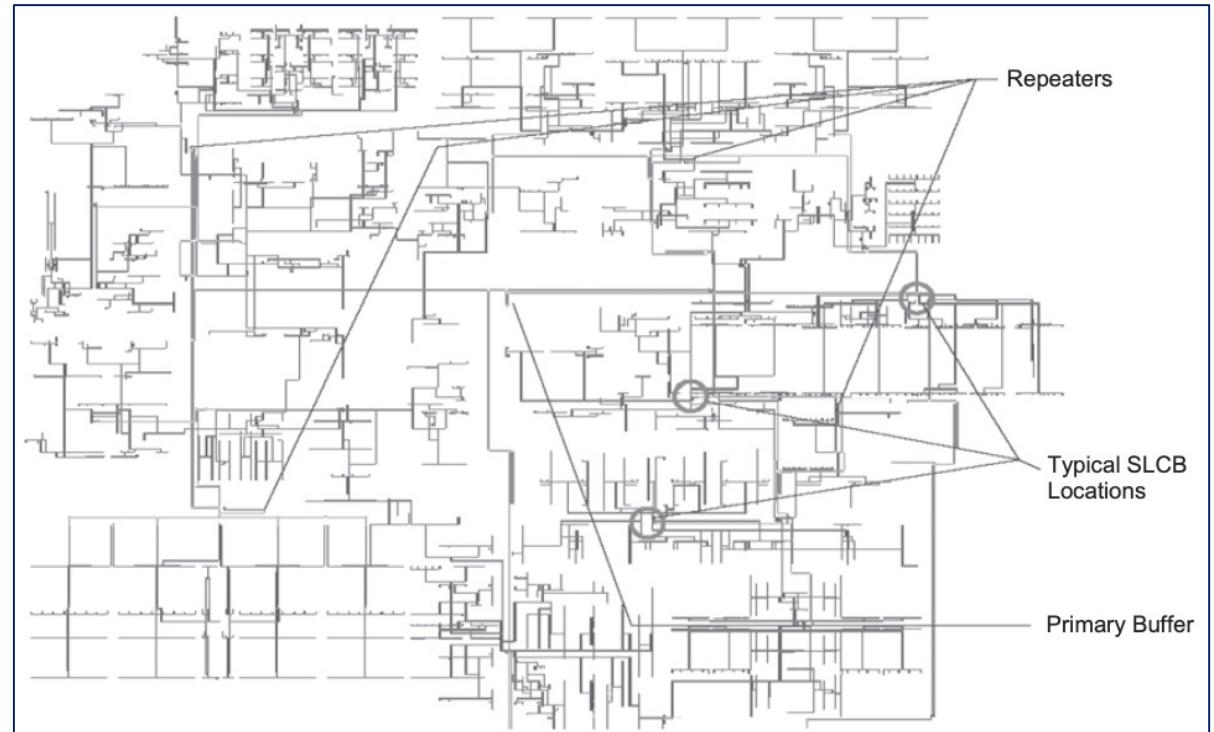
- ◆ Fractal structure
 - Gets clock arbitrarily close to any point
 - Matched delay along all paths
- ◆ Delay variations cause skew
- ◆ A and B might see big skew



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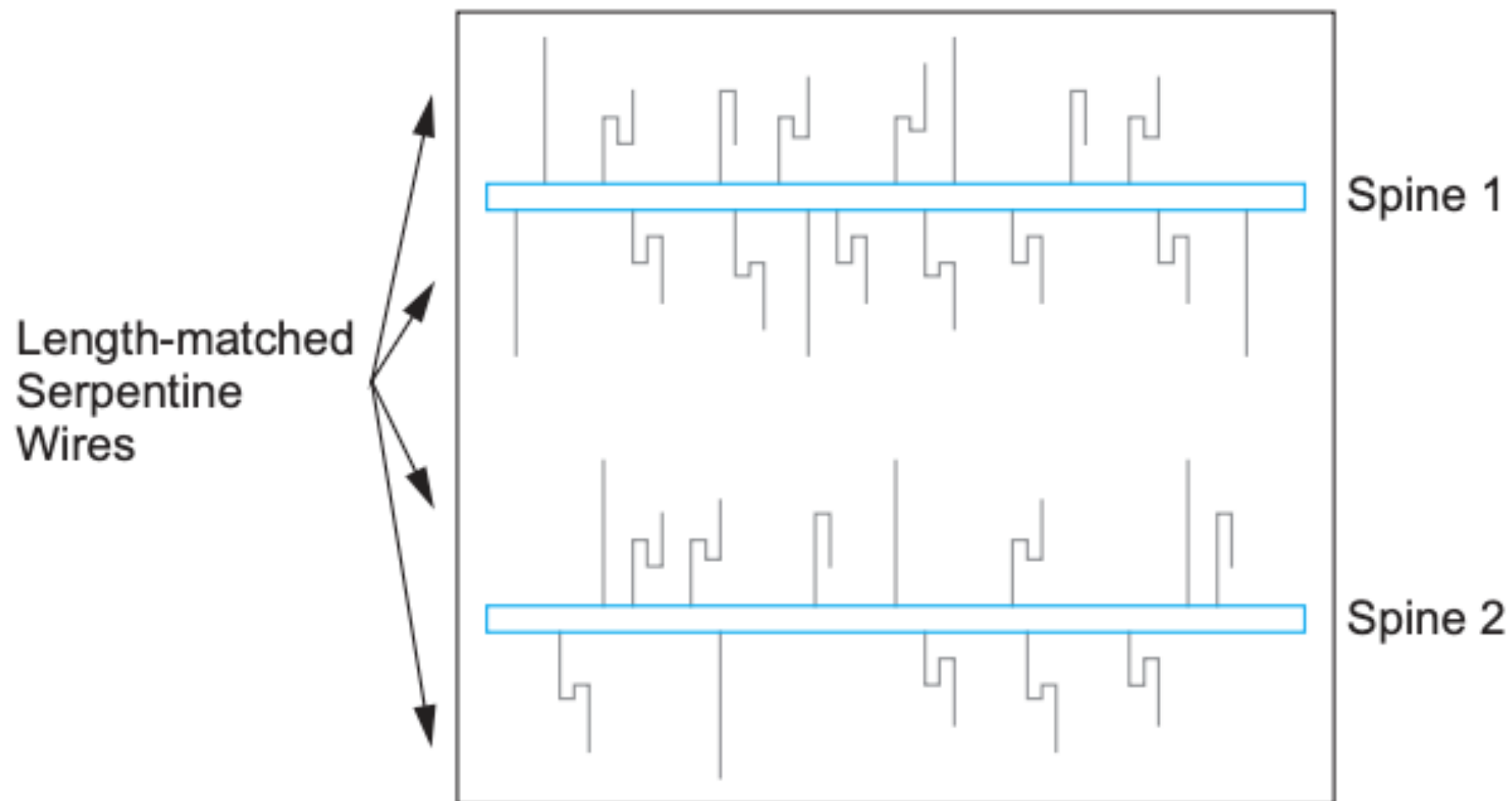
Practical H-Tree network

- ◆ Itanium 2 has four levels of buffering:
 1. Primary driver
 2. Repeater
 3. Second-level clock buffer
 4. Gater
- ◆ Route around obstructions
- ◆ IBM Power4, PowerPC
 - H-tree drives 16-64 sector buffers
 - Buffers drive total of 1024 points
 - All points shorted together with grid



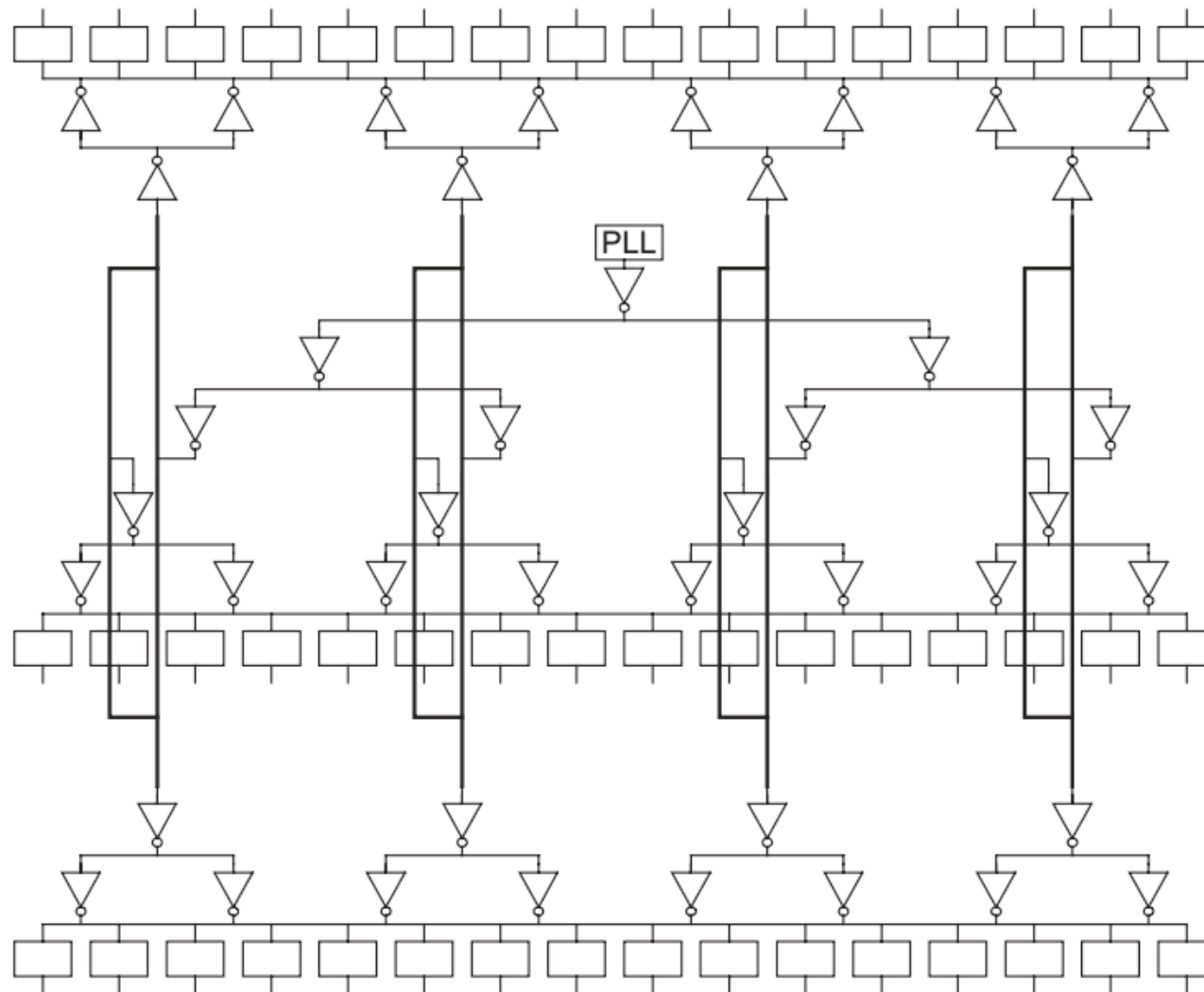
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Clock spines with Serpentine Routing

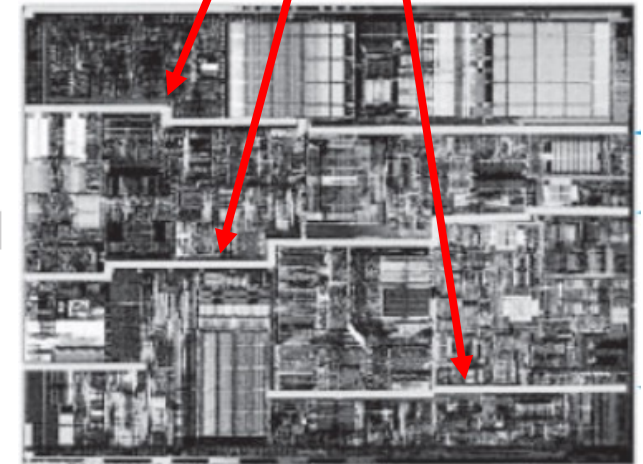


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Pentium 4 has 3 clock spines



clock spines



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